

NEXORY – Refined Tech Stack & Free Tier Breakdown

1. Auth & Permission

Choice: Supabase Auth

Free Tier: 50,000 users, unlimited API requests

Why: Built-in Row Level Security (RLS) directly supports controlled access to user-saved content while keeping MVP development simple and cost-effective.

2. Nexory App / PWA

Choice: React + Vite + TailwindCSS

Free Tier: Vercel / Netlify – 100GB bandwidth

Why: The PWA approach provides a native-like mobile experience without App Store costs.

3. Backend API

Choice: FastAPI (Python)

Free Tier: Render / Fly.io – 512MB RAM, 3GB storage

Why: FastAPI supports OCR processing and embedding generation efficiently while simplifying AI integration.

4. Database

Choice: Supabase PostgreSQL

Free Tier: 500MB database

Why: Keeps authentication, metadata, and vector search support inside one unified environment.

5. File Storage

Choice: Supabase Storage

Free Tier: 1GB free storage

Why: Simplifies screenshot and uploaded-content handling while remaining tied to authentication.

6. Vector Search

Choice: pgvector

Free Tier: Included in Supabase

Why: Supports lightweight semantic retrieval without requiring additional vector database vendors.

7. OCR Processing

Choice: Tesseract + PyTesseract

Free Tier: Self-hosted – \$0

Why: Enables screenshot text extraction without expensive cloud OCR services.

8. AI Understanding

Choice:

- Embeddings: BAAI/bge-small-en
- Summaries & Tags: Llama 3.1 8B via Groq

Free Tier: 1M tokens/day**Why:** Enables fast semantic understanding and retrieval while reducing API costs.

9. Voice Search

Choice: Web Speech API**Free Tier:** Browser-native – \$0**Why:** Allows users search naturally using voice without additional backend cost.

10. Translation

Choice: LibreTranslate**Free Tier:** Self-hosted – \$0**Why:** Supports multilingual query normalization during future scaling.

11. Retrieval Engine

Choice: pgvector cosine similarity + keyword fallback**Free Tier:** Included**Why:** Enables meaning-based retrieval instead of relying only on exact keyword matching.

12. Frontend Hosting

Choice: Vercel (Hobby)**Free Tier:** 100GB bandwidth, 1000 build minutes**Why:** Provides automatic GitHub deployment and optimized frontend delivery.

Estimated MVP Cost

Total Monthly Cost at MVP Scale (500 Users):

\$0 during MVP stage, with costs only increasing if storage or AI free-tier limits are exceeded.

Core Technical Direction

Nexory's architecture prioritizes:

- intelligent retrieval
- semantic search
- scalable AI memory systems
- low-cost MVP deployment
- fast iteration during The Build